

Deterministic Optimization

Linear Optimization Modeling
Decision-Making under
Uncertainty

Andy Sun

Assistant Professor

Stewart School of Industrial and Systems Engineering

The Need to Make Decision Under
Uncertainty

Modeling using Linear Programs

Learning Objectives for this Lesson

- Study motivating examples for decision making under uncertainty
- Discover the key decision structure for decision making under uncertainty

Inventory Control Problem

- A company designs, produces, and sells winter fashion items such as ski jackets.
- The company must commit to specific production quantity x before knowing the exact demand d , 3 months before the winter season.
- Demand d is estimated as a random variable.
- After seeing demand, the company decides how many to sell and how many to sell at a discounted price v .
- This is called *Decision Making under Uncertainty*, because decision x is made under uncertain demand d .



Generation Capacity Expansion

- An electric utility company plans to build new generation stations to serve growing demand, called generation capacity expansion.
- New generation capacity has to be decided before demand and future fuel prices are known.
- Future demand and fuel prices are not known at the moment of making capacity decision, but can be estimated as random variables.
- After demand is realized, the utility company schedules existing and new generators based on capacity expansion decision.
- Again, this is a problem of decision making under uncertainty.



Financial Planning

- A family wishes to provide for a child's college education 12 years later.
- The family currently has \$100K and decides how to invest in any of 5 investments, e.g. stocks, bonds, etc.
- Investment can be adjusted every 4 years. So there are 3 investment periods.
- The returns of investments are unknown and modeled as random variables.
- The family wants to maximize the total expected return.
- A problem of decision making under uncertainty.



Decision Types

- In each example, some decisions (production, capacity expansion, initial investment) have to be made before uncertainty (ski-jacket demand, future electricity demand, fuel prices, investment returns) is known.
- Some other decisions (sell/discount quantity, generation scheduling, investment adjustment) can be made after uncertainty is realized.
- First type decision is called *Here-and-Now decision*, made before knowing uncertain parameters.
- Second type decision is called *Wait-and-See* decision, made after knowing uncertain parameters.
- We will learn stochastic programs to implement this decision structure.

Summary

- We discussed 3 examples with significant uncertainty (demand, prices, returns etc).
- Decision making under uncertainty is necessary due to lack of information (e.g. future demand), or difficulty or impossibility to forecast accurately (e.g. stock markets, weather).
- Two-types of decisions
 - Here-and-Now
 - Wait-and-See